Knowledge Based Systems — Complete Lecture Notes

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Lecture 1 — Course Overview & Introduction to Knowledge Based Systems

1.1 Convergence: From Data to Intelligence

The path from raw data to wisdom follows a hierarchy, with each level adding context and structure:

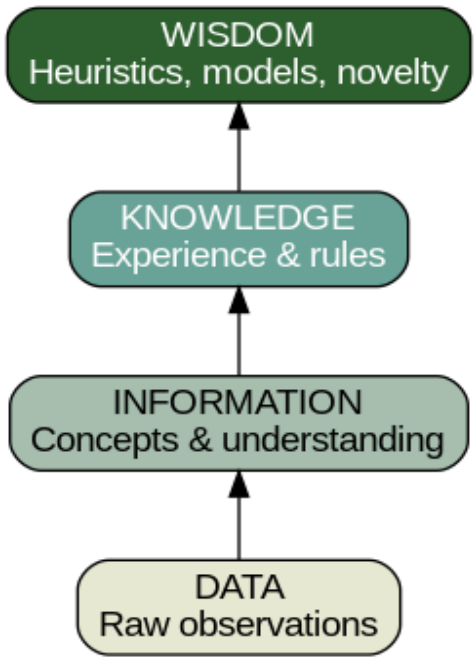


Diagram 1

Level	Description
Data	Raw observations gathered through fact finding
Information	Data placed in context (concepts, understanding)
Knowledge	Information structured through experience and rules
Wisdom	Heuristics, models, and novelty applied to knowledge

1.2 What is a Knowledge-Based System?

A Knowledge-Based System (KBS) is an attempt to represent knowledge explicitly together with a reasoning system that allows it to derive new knowledge.

Every KBS has two distinguishing features:

- **Knowledge Base** — explicit representation of domain facts and rules.
- **Inference Engine** — reasoning mechanism that derives new conclusions.

Diagram 2

1.3 Information Systems Hierarchy

Computer-based systems scale with the volume of data and the sophistication of reasoning:

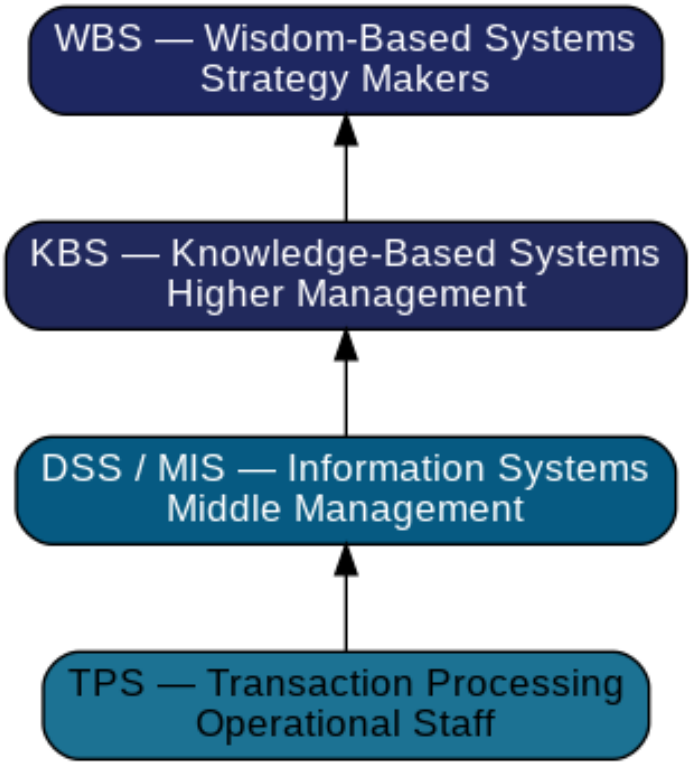


Diagram 3

Layer	System Type	Purpose
Wisdom	WBS (Wisdom-Based Systems)	Strategy makers apply morals, principles, and experience to generate policies
Knowledge	KBS (Knowledge-Based Systems)	Higher management generates knowledge by synthesizing information
Information	DSS / MIS	Middle management uses analysis-derived reports to act
Data	TPS (Transaction Processing)	Operational staff perform basic transactions

1.4 Business Intelligence Systems

Business Intelligence (BI) systems and analytical software applications read data, process it, and extract and report useful insights for decision-making.

1.5 Assessment Breakdown

Assessment	Weighting
Group Project	30%
Mid Term Exam	20%
Final Exam	50%
Total	**100%**

Lecture 2 — Big Data and Data Analytics

2.1 Fields that Generate Data

- **Social networking sites** — information, posts, and links from platforms like Facebook and Twitter.
- **Search engines** — vast data retrieved from different databases.
- **Medical history** — patient health records from hospitals.
- **Online shopping** — customer preferences across products.
- **Stock exchange** — shares of different companies held by stockholders.

2.2 What is Big Data?

Big Data is a collection of large datasets that cannot be adequately processed using traditional processing techniques. It describes the *volume* of structured and unstructured data produced in day-to-day business environments.

2.3 Characteristics of Big Data (The V's)

Key dimensions typically used to describe Big Data:

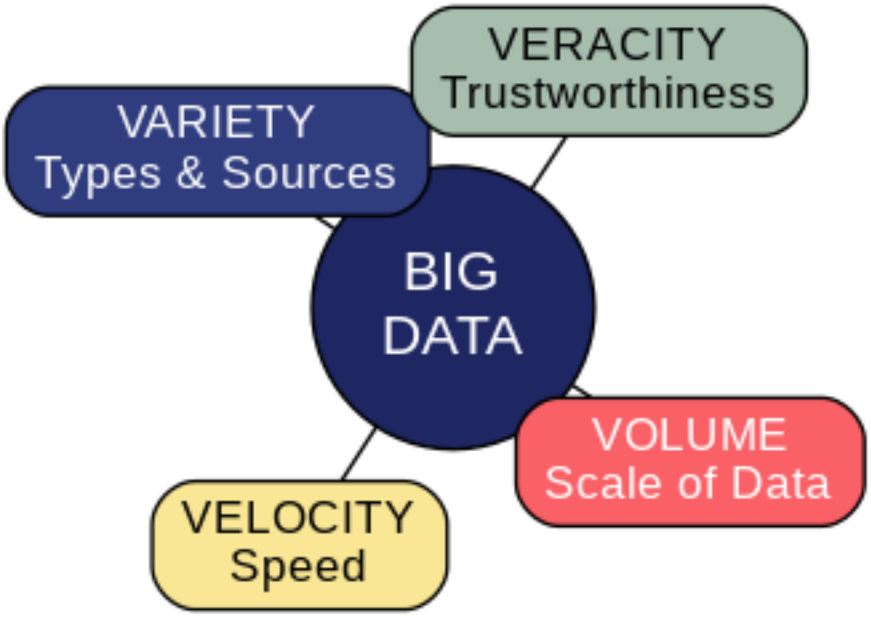


Diagram 4

- **Volume** — the scale of data.
- **Velocity** — the speed at which data is generated and processed.
- **Variety** — the diversity of data types and sources.
- **Veracity** — covers origin, authenticity, trustworthiness, integrity, and completeness.

2.4 Categories of Big Data

Category	Description
Black Box Data	Conversations between crew members and alert/communication messages
Social Media Data	Information and views posted by millions worldwide
Stock Exchange Data	Details of business transactions in and out
Power Grid Data	Information consumed per node (e.g., a base station)
Transport Data	Model, capacity, distance, and availability of vehicles
Search Engine Data	Large-scale data retrieved from multiple databases

2.5 Traditional Data vs. Big Data

Traditional data is typically structured, smaller-scale, and handled by relational database systems. Big Data is large-scale, frequently unstructured or semi-structured, and requires distributed, parallel processing frameworks.

2.6 Big Data Analytics

A method to uncover hidden patterns in large data and extract useful information. It involves inspecting, differentiating, and transforming big data in order to:

- Identify useful information.
- Suggest conclusions.
- Guide accurate decision-making.

Big data analytics spans data mining and communication, and supports decision making.

2.7 Big Data Analytics Workflow

Diagram 5

1. **Data Extraction**
2. **Data Integration**
3. **Big Data Computing & Data Management**
4. **Data Mining & Visualization**
5. **Applications**

2.8 Analytics Models and Technologies

Big data technologies are divided into two broad categories:

- **Analytics on data at rest** — batch processing of stored datasets.
- **Analytics on data in motion** — streaming/real-time analytics.

Multi-disciplinary big-data analytics pursues interdisciplinary, multi-university, and multi-team research on heterogeneous scientific and technological data.

2.9 Tips for Creating Effective Big Data Models

- **Don't impose traditional modeling techniques on big data.** New data sources and forms may emerge at any time.
- **Design a system, not a schema.** Big data systems may use schemaless stores like NoSQL. Models should cover business information requirements, governance, storage, integration, and variety of data types.
- **Look for big data modeling tools.** Decision-makers should explicitly require modeling capability for big data.
- **Focus on data that is core to your business.** Don't model everything — model only what is essential.
- **Deliver quality data.** The more you know about each piece of data, the better you can place it in the model.
- **Look for key inroads into the data.** A common vector today is geographical location; such entry points help design access paths.

2.10 Big Data Challenges and Solutions

Challenge	Description
Lack of knowledgeable professionals	Shortage of data scientists, analysts, and engineers
Lack of understanding of massive data	Employees may not grasp data's storage, processing, importance, and sources
Data growth issues	Exponential growth, mostly unstructured, is hard to handle
Tool selection confusion	Choosing between HBase/Cassandra or Hadoop MapReduce/Spark is not obvious
Integrating data from a spread of sources	Data comes from many systems; unification is crucial for reporting and BI
Securing data	Breaches can cost organizations up to $3.7 million per incident
High cost of data and infrastructure	50% of US and 39% of European executives cite limited IT budgets as the biggest barrier

2.11 Barriers in Big Data

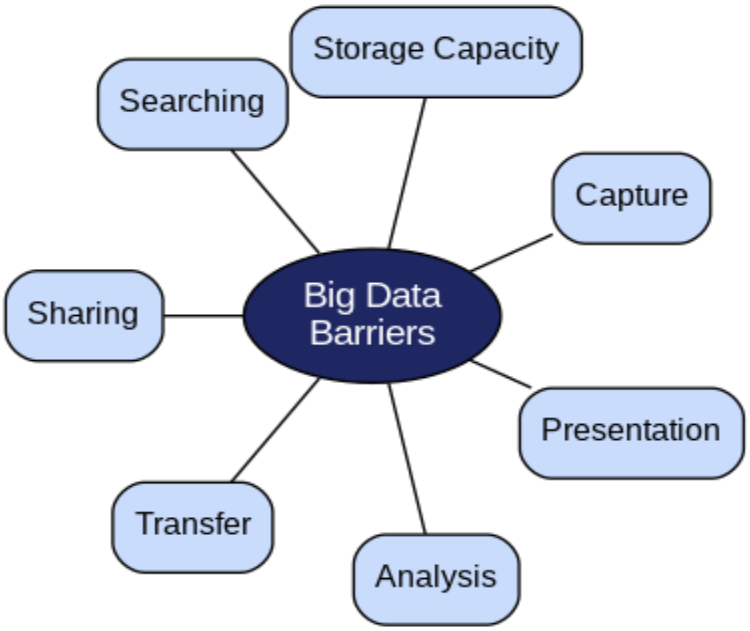


Diagram 6

Capture, storage capacity, searching, sharing, transfer, analysis, and presentation.

2.12 Importance of Big Data Analytics

Big data analytics enables in-depth analysis for better decisions and strategy. It supports:

- Finding causes of failures, issues, and defects in real time.
- Recalculating entire risk portfolios in minutes.
- Detecting fraudulent behavior before it affects the organization.
- Discovering new revenue opportunities.
- More effective marketing and better customer service.
- Improved operational efficiency and competitive advantage.

Lecture 3 — Data Visualization

Data visualization is the process of creating graphical representations of information.

Diagram 7

3.1 Common Visualization Techniques

The following table summarizes the principal visualization techniques covered in the lecture:

Technique	Best Used For	Key Strengths / Caveats
Pie Chart	Part-to-whole proportions	Simple for general audiences; poor for complex data
Bar Chart	Comparing categories to a measured value	Clear; can be cluttered with many categories
Histogram	Distribution over a continuous interval	Shows frequency; reveals concentrations and gaps
Gantt Chart	Project timelines and tasks	Tracks every aspect of a project
Heat Map	Differences via color variation	Needs clear legend; reveals trends quickly
Box and Whisker Plot	Quartile-based data summary	Shows median, outliers, skew
Waterfall Chart	How a value changes over time	Popular for spending/earnings over time
Area Chart	Quantities over time; part-to-whole	Stacked areas compare multiple series
Scatter Plot	Relationships between variables	Strong for large datasets; shows correlation
Pictogram Chart	Simple, engaging presentations	Icons bridge language/cultural gaps
Timeline	Chronological sequences	Highlights key events and patterns
Highlight Table	Comparing categorical data	Conditional formatting via colors
Bullet Graph	Performance vs. benchmarks	Shows actual vs. target across ranges
Choropleth Map	Values across geographic regions	Color progression; optional interactivity
Word Cloud	Frequency in textual data	Word size reflects frequency
Network Diagram	Relationships between qualitative points	Nodes and edges show connections
Correlation Matrix	Patterns among many variables	Color-scaled cells reveal correlations

3.2 Technique Details

Pie Chart

Ideal for illustrating proportions, or part-to-whole comparisons. Best for audiences unfamiliar with the detailed information or interested only in key takeaways. Falls short for thorough explanations of complex information.

Bar Chart

One axis shows the categories compared; the other shows a measured value. Bar length indicates how each group measures against the value. Too many categories make labeling problematic; may be too simple for complex datasets.

Histogram

Illustrates the distribution of data over a continuous interval or defined period. Useful for identifying where values are concentrated, where there are gaps, or where unusual values occur, and for showing frequency.

Gantt Chart

Useful in illustrating a project timeline or task progression. Allows team members to track every aspect of a project and stay organized.

Heat Map

Shows differences in data through variations in color. A clear legend is required so that users can successfully read and interpret the heatmap. For example, a heat map could map day-of-week vs. time-of-day shaded by sales volume to reveal peak periods.

Box and Whisker Plot

Summarizes data via quartiles: a box from the first to the third quartile, a line inside marking the median, and whiskers extending to the minimum and maximum. Outliers appear as individual points. Helpful for quickly determining whether data is symmetrical or skewed.

Waterfall Chart

Illustrates how a value changes as it is influenced by different factors (often time). Popular for visualizing spending or earnings over time.

Area Chart

A variation of a line graph where the area beneath the line is shaded to represent total value. Stacked area charts compare several data series on the same graph; excellent for part-to-whole comparisons over time.

Scatter Plot

Useful for illustrating relationships between variables and for identifying trends or correlations. Most effective for fairly large datasets — the tighter the cluster, the stronger the correlation.

Pictogram Chart

Uses icons to visualize data, each icon representing a value or category. Works well when language or cultural differences could hinder audience comprehension.

Timeline

The most effective way to visualize a sequence of events in chronological order, often used for historical or future-planning information. Timelines become more engaging with images, colors, fonts, and decorative shapes.

Highlight Table

Makes it easier to spot trends and patterns in categorical data by applying conditional formatting rules that color cells meeting given conditions.

Bullet Graph

Informs the viewer how a business performs against benchmarks for key metrics. A dark horizontal bar represents the actual value, a vertical line is the target, and colored segments represent range scores such as “poor,” “fair,” or “good.”

Choropleth Maps

Uses color and shading to visualize numerical values across geographic regions. Color progression shows variation between regions; some tools add interactivity to show exact values.

Word Cloud

Visual representation of text where word size is proportional to frequency in the dataset. Useful for qualitative data analysis and keyword identification.

Network Diagram

Represents relationships between qualitative data points. Composed of **nodes** (data points) and **links / edges** (relationships between nodes).

Correlation Matrix

Each cell represents the relationship between two variables, with color scale indicating the strength of correlation. Useful for summarizing and finding patterns in large datasets.

3.3 Tips for Effective Visualizations

- **Evaluate your audience** before choosing complexity and style.
- **Avoid unnecessary distractions**; keep the focus on the data.
- **Be mindful of color and design** — low contrast makes differentiation hard, while overly bold palettes overwhelm.

3.4 Common Mistakes to Avoid

- **Using the wrong type of chart or graph.** Match chart type to data type (qualitative vs. quantitative).
- **Including too many variables.** Include only relevant, essential information.
- **Using inconsistent scales.** Inconsistent scales cause viewer confusion.
- **Unclear linear vs. logarithmic scaling.** On a linear scale, the distance between any two values is equal; on a log scale it varies in a specific pattern. Mixing them up may understate significance.
- **Poor color choices.** Avoid too many colors, surprising uses of familiar colors (e.g., red/green), low contrast, and failing to account for colorblind viewers.

3.5 Popular Data Visualization Tools

Tool	Strengths
Microsoft Excel / Power BI	Ubiquitous; Excel covers ~20 chart types; Power BI is purpose-built for analytics and visualization
Google Charts	Free; integrates with Salesforce, SQL, Sheets; HTML5/SVG output; 18+ chart types
Tableau	Easy yet powerful; connects to hundreds of sources; desktop, server, web, and free Public versions
Zoho Analytics	Tailored to business intelligence (sales, marketing, revenue pipelines); user-friendly dashboards
Datawrapper	Journalism-friendly; simple but limited (manual data entry); good for charts, maps, and web graphics
Infogram	Charts, reports, maps, and infographics; drag-and-drop editor; exports to image, GIF, HTML

Lecture 4 — Knowledge Representation

Knowledge representation techniques covered: **Rules**, **Semantic Networks**, **Frames**, and **Formal Logic**.

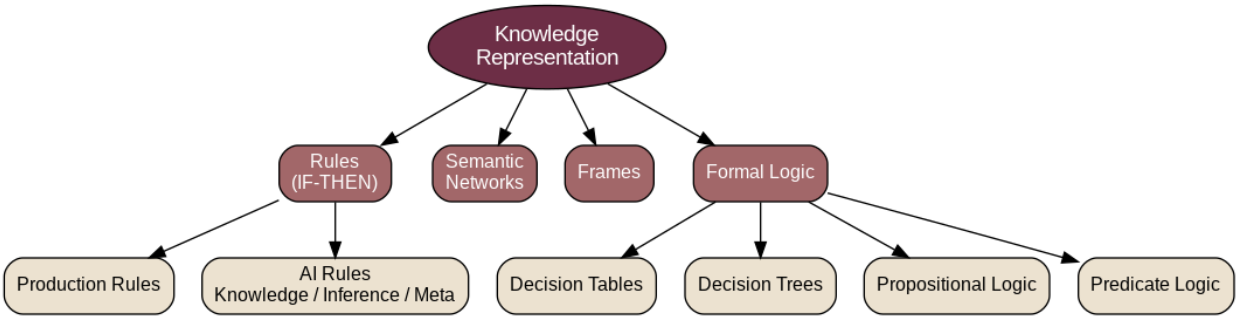


Diagram 8

4.1 Production Rules (IF-THEN)

Production rules are independent pieces of knowledge that combine to produce results. They model human behavior and follow patterns like:

- `IF condition THEN conclusion`
- `conclusion IF condition`
- `IF condition THEN conclusion1 ELSE conclusion2`

Advantages: simple syntax, easy to understand, simple interpreter, highly modular, flexible (easy to add or modify).

Disadvantages: hard to follow hierarchies, inefficient for large systems, not all knowledge is rule-expressible, poor at structured descriptive knowledge.

4.2 Types of AI Rules

- **Knowledge rules** — declare facts and relationships; stored in the knowledge base.
- **Inference rules** — advise how to proceed given facts; part of the inference engine.
- **Metarules** — rules about how to use other rules.

4.3 Inference Engine

Algorithms that direct the search through a knowledge base:

- **Forward chaining** — data driven; start with information, draw conclusions.
- **Backward chaining** — goal driven; start with expected conclusions, seek supporting evidence.
- **Inference/goal tree** — schematic view with AND/OR/NOT nodes; answers "why" and "how" questions.

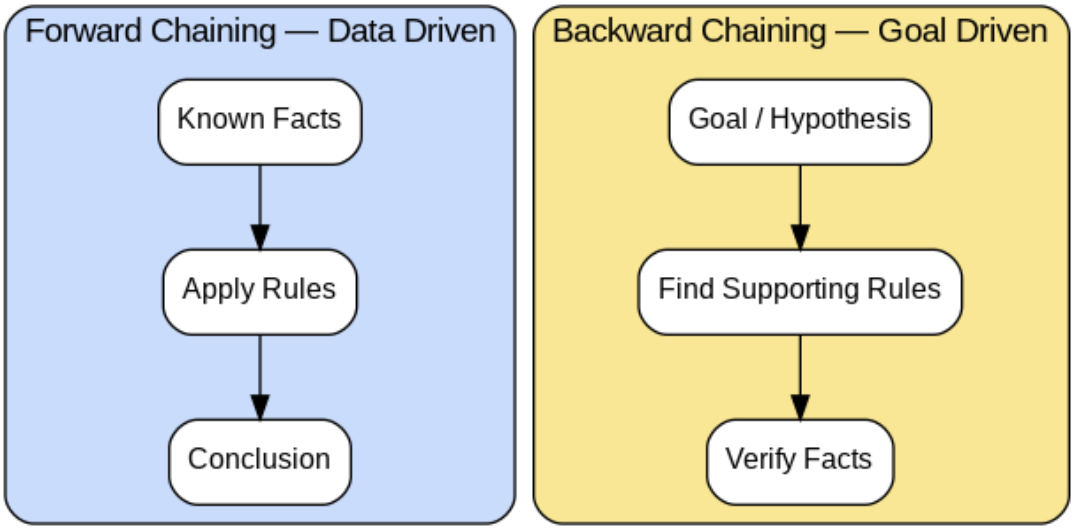


Diagram 9

4.4 Advantages and Limitations of AI Rules

Advantages: easy to understand, modify, and maintain; easy explanations; independent rules; uncertainty easily combined with rules.

Limitations: huge numbers of rules may be required; designers may force knowledge into rule form; search limitations and evaluation difficulties.

4.5 Semantic Networks

Graphical depictions made of **nodes** and **links** expressing hierarchical relationships between concepts, with built-in support for inheritance.

Example associations: "Adel *is a* Boy", "Boy *is a* Human being", "Adel *goes to* School", "Ahmed *works for* ACME", "BMW *made in* Germany".

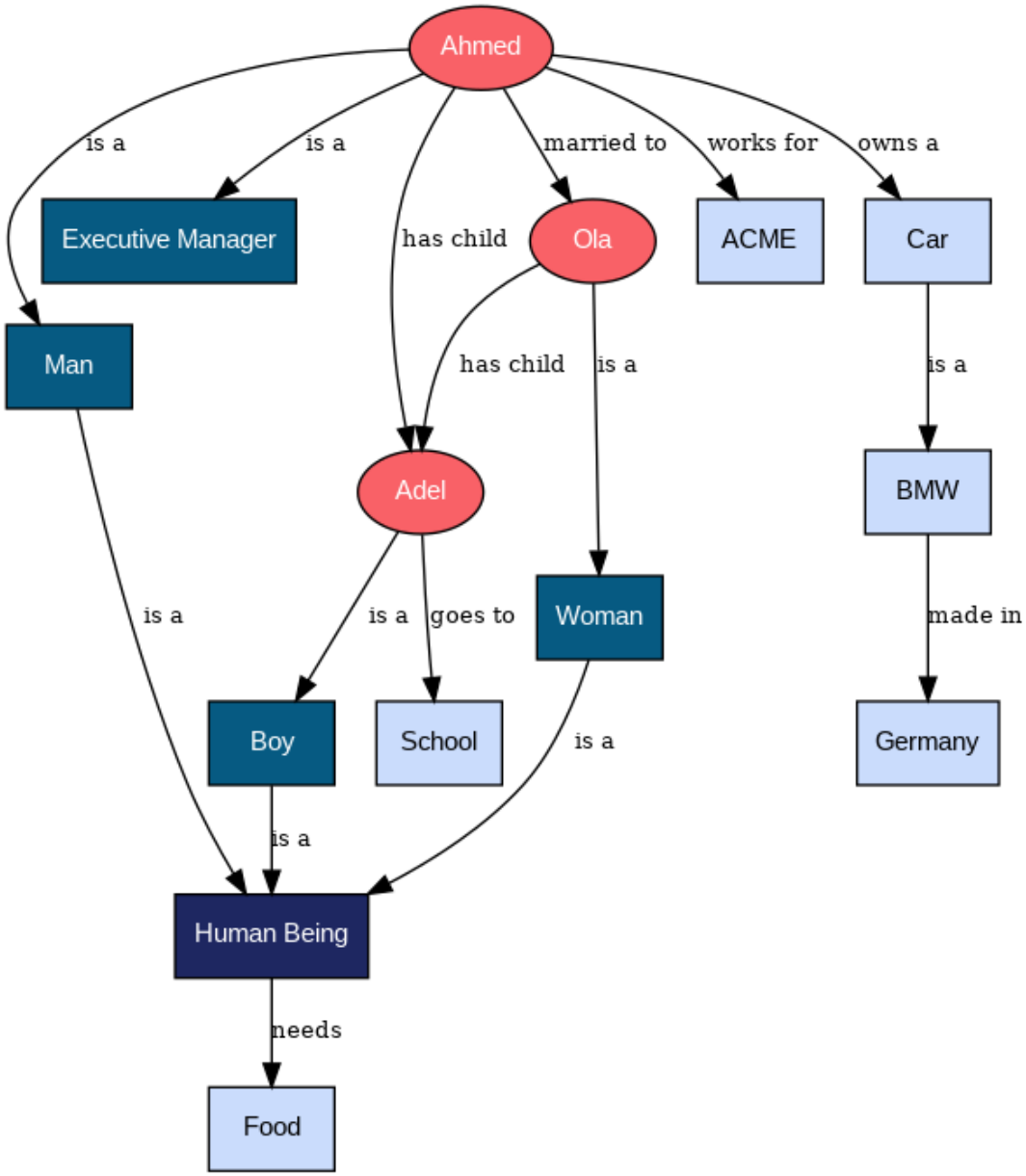


Diagram 10

Advantages: easy to follow hierarchy, trace associations, and remain flexible.

Disadvantages: node meanings can be ambiguous; difficult exception handling; hard to program.

4.6 Frames

Frames hold all knowledge about an object in a hierarchical structure that supports inheritance.

- Based on object-oriented programming principles.
- Organize knowledge by characteristics and attributes.
- Use **slots** and **subslots/facets**.
- Parents hold general attributes, instantiated to children.
- Often combined with production rules.

Advantages: expressive power; easy to set up; slots support new properties and relations; easy to create specialized procedures; default values and missing-value detection are built in.

Disadvantages: difficult to program and reason with; lack of inexpensive software.

4.7 Formal Logic

Formal logic representations include:

- **Decision tables** — spreadsheet format where all possible attributes are compared to conclusions.
- **Decision trees** — nodes and links used for knowledge diagramming.
- **Computational logic**
  - *Propositional logic* — true/false statements.
  - *Predicate logic* — variable functions applied to components of statements, based on calculus (e.g., `J = Passed assignment`, `K = Passed exam`, `Z = J AND K`).

Advantages: facts asserted independently of use; ensures only valid consequences (precision); completeness.

Disadvantages: separation of representation and processing; inefficient on large data sets; very slow with large knowledge bases.

4.8 Explanation Facility

The explanation facility (or *justifier*) makes the system more understandable by:

- Exposing shortcomings.
- Explaining unanticipated situations.
- Satisfying the user's psychological and social needs.
- Clarifying underlying assumptions.
- Conducting sensitivity analysis.

Types of explanations: Why, How, Journalistic (who, what, where, when, why, how), and Why not.

4.9 Generating Explanations

- **Static explanation** — pre-inserted text.
- **Dynamic explanation** — reconstructed via rule evaluation.
- **Tracing** — records or line of reasoning.
- **Justification** — based on empirical associations.
- **Strategic use of metaknowledge.**

4.10 Handling Uncertainty

Uncertainty is widespread and an important KBS component. Representations include:

- **Numeric scale** — e.g., 1 to 100.
- **Graphical presentation** — bars, pie charts.
- **Symbolic scales** — "very likely" to "very unlikely".

Formal approaches:

- **Probability ratio** — degree of confidence; chance of occurrence.
- **Bayes Theory** — subjective probabilities for propositions; imprecise; combines values.
- **Dempster–Shafer** — belief functions with boundaries on assigned probabilities; assumes statistical independence.
- **Certainty factors** — represent belief in an event given evidence. Belief and disbelief are independent and not directly combinable; certainty factors can be combined within a rule, and multiple rules can be combined.

4.11 Evaluation, Validation, and Verification

- **Evaluation** — assesses the system's overall value.
- **Validation** — compares system performance to experts; identifies concordance and differences.
- **Verification** — ensures the system is built and implemented correctly; often automatable.

4.12 Measures of Validation

Measure	Meaning
Accuracy	How well the system reflects reality; correctness of the knowledge base
Adaptability	Possibility of future development and changes
Adequacy (Completeness)	Portion of necessary knowledge included
Breadth	How well the domain is covered
Depth	Degree of detailed knowledge
Face validity	Credibility of the knowledge
Generality	Capability to be used with similar problems
Precision	Ability to replicate parameters; consistency of advice

Realism	Accounts for relevant variables; similarity to reality
Reliability	Fraction of predictions empirically correct
Robustness	Sensitivity of conclusions to model structure
Sensitivity	Impact of knowledge-base changes on output quality
Technical/Operational validity	Quality of assumptions; context constraints
Turing test	Can a human evaluator tell system conclusions from expert's?
Usefulness	Adequacy of knowledge for correct problem solving
Validity	Capability to produce empirically correct predictions

Lecture 5 — Knowledge Engineering

5.1 Definition

Knowledge engineering is the process of **eliciting**, **structuring**, **formalizing**, and **operationalizing** the information and knowledge involved in a knowledge-intensive problem domain, in order to construct a program that can perform a difficult task adequately.

It is not simply "mining from the expert's head" — it consists of constructing different aspect-models of human knowledge. Multiple representations exist: textbooks, graphical representations, heuristics, and skills.

5.2 Why Proper Knowledge Engineering Matters

- Knowledge is valuable and often outlives a particular implementation → knowledge management.
- Errors in a knowledge base can cause serious problems.
- Systems face heavy demands on extendibility and maintenance as knowledge changes over time.

5.3 Methodological Pyramid

From bottom to top (each level feeds back into the lower layers):

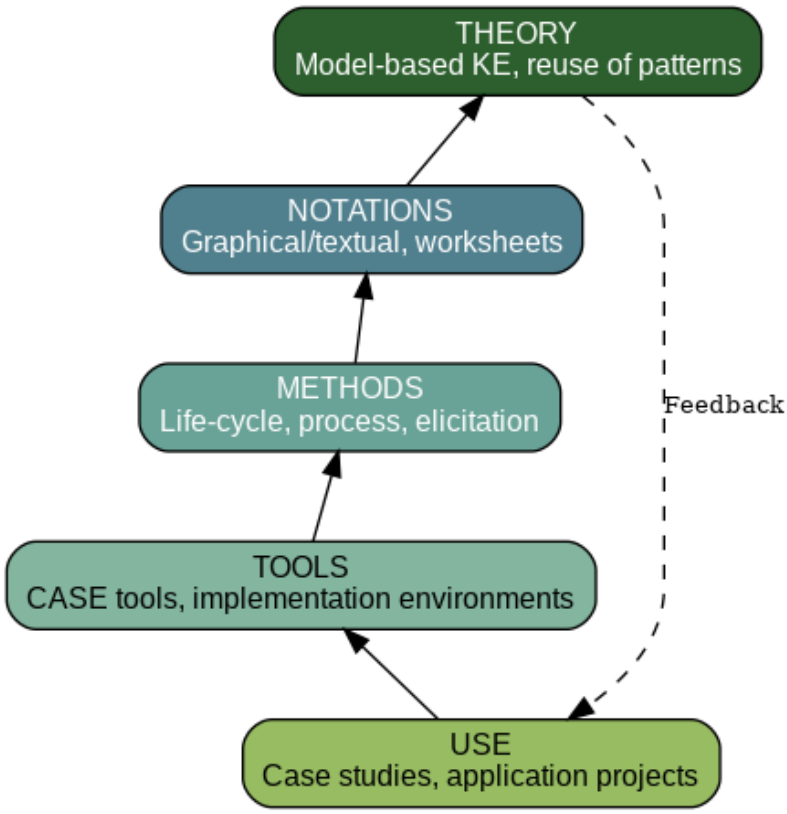


Diagram 11

1. **Theory** — model-based knowledge engineering; reuse of knowledge patterns.
2. **Methods** — graphical/textual notations, worksheets, document structures.
3. **Tools** — life-cycle model, process model, guidelines, elicitation techniques.
4. **Use** — case studies, application projects, CASE tools, implementation environments.
5. **Feedback** — continuous loop back into theory and methods.

5.4 Model-Based Knowledge Engineering

- Each model emphasizes certain aspects and abstracts from others.
- Models decompose knowledge-engineering tasks: while building one model, the engineer can neglect other aspects.
- Concentrate on the conceptual structure of knowledge first; leave programming details for later.
- Knowledge has a stable internal structure analyzable by distinguishing specific knowledge types and roles.

5.5 The Model Set

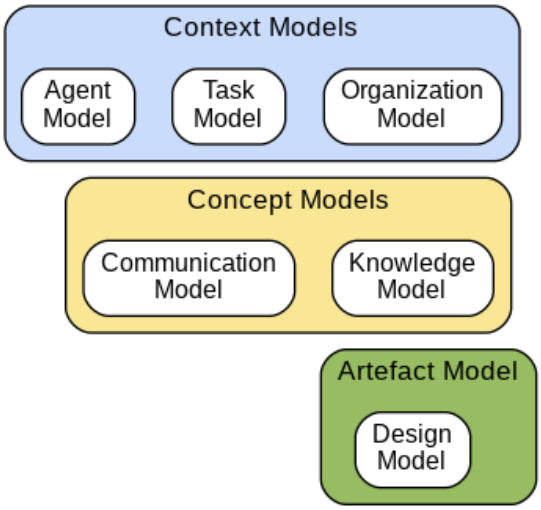


Diagram 12

Model	Role
Organization model	Analyzes the organization; discovers problems, opportunities, and KBS development impacts
Task model	Describes tasks currently performed or to be performed
Agent model	Describes capabilities, norms, preferences, and permissions of agents (task executors)
Knowledge model	Implementation-independent description of knowledge involved in a task
Communication model	Models communicative transactions between agents
Design model	Describes the structure of the system to be constructed

5.6 Principles of the Model Set

- Divide and conquer.
- Configure an adequate model set per application.
- Models evolve through well-defined states.
- The model set supports project management.
- Model development is driven by project objectives and risk.
- Models can be developed in parallel.

5.7 Forms of Models

- **Model template** — predefined, fixed, configurable structure.
- **Model instance** — objects manipulated during a project.
- **Model versions** — versions of a model instance.
- **Multiple model instances** — separate instances (e.g., "current" vs. "future" organization).

5.8 The Product

The final product of a knowledge engineering effort consists of:

- **Instantiated models** — represent the environment and the delivered KBS.
- **Additional documentation** — information not represented in model templates (e.g., project management data).
- **Software** — the running system itself.

5.9 Roles in Knowledge-System Development

Role	Main Responsibilities
Knowledge Provider / Specialist	Traditional expert with extensive domain experience; advises beginners where to start; inter-provider differences are common
Knowledge Engineer	Mediates between expert and knowledge base; elicits, refines, and represents knowledge; interprets and integrates answers; poses counterexamples
Knowledge-System Developer	Implements the system on a target platform; needs design/implementation expertise; understands knowledge analysis at the "use" level; often the same person as the knowledge engineer
Knowledge User	Primary users interact directly with the system; secondary users are affected indirectly; skill level and attitude shape explanation needs
Project Manager	Plans, schedules, and monitors development (typically 4–6 person projects); communicates with clients
Knowledge Manager	Background role; monitors organizational purpose of systems and knowledge assets; initiates follow-up projects; supports reuse and team setup

5.10 Terminology

- **Domain** — area of interest (banking, food industry, photocopiers, car manufacturing).
- **Task** — something that needs to be done by an agent (monitoring, planning, analyzing).
- **Agent** — executor of a task (human or software system).
- **Application task** — top-level task to be performed in a given application.

5.11 Issues in Developing Knowledge-Based Systems

- Requirements are harder to define than for standard software.
- Human expertise is difficult to define and elicit; close continuous contact with experts is required throughout development.
- Desirable qualities of knowledge engineers include intuition, broad/deep absorption of non-CS knowledge, and interpersonal skills — alongside standard software engineering skills (organization, documentation, debugging, testing).

Lecture 6 — Knowledge and Knowledge Acquisition

6.1 What is Knowledge?

Knowledge includes **facts, concepts, procedures, models, heuristics,** and **examples**. It can be:

- Specific or general.
- Exact or fuzzy.
- Procedural or declarative.

6.2 Knowledge and Uncertainty

There is often some degree of uncertainty in knowledge — things are not always strictly true or false, and knowledge may be incomplete. In an expert system, **certainty factors** indicate the degree of certainty attached to a fact or rule.

6.3 Data Volume and Computer-Based Systems

As the volume and sophistication of reasoning grow, the type of system shifts accordingly:

Level	Users	System
Wisdom	Strategy makers applying morals, principles, and experience	WBS
Knowledge	Higher management synthesizing information	KBS
Information	Middle management using analysis-derived reports	DSS, MIS
Data	Operational staff performing basic transactions	TPS

6.4 Sources of Knowledge

- **Documented** — books, journals, procedures, films, databases.
- **Undocumented** — people's knowledge and expertise, stored in their minds and other senses.

6.5 Knowledge Source Declaration

- **Explicit knowledge** — structured and documented (computer files, databases, policy/procedure manuals).
- **Tacit knowledge** — expertise and experience not formally documented.
- **Organizational memory** — stored learning from an organization's history, used for decisions and reference.

6.6 Levels of Knowledge

- **Shallow knowledge** — specific to a situation; limited IF-THEN rules; little meaning; no explanation.
- **Deep knowledge** — supports problem solving with an internal causal structure; built from many inputs including emotions, common sense, and intuition; difficult to encode.

6.7 Categories of Knowledge

- **Declarative** — descriptive, factual, shallow knowledge.
- **Procedural** — how things work; supports inference.
- **Semantic** — symbolic knowledge.
- **Episodic** — autobiographical, experiential.
- **Meta-knowledge** — knowledge about knowledge.

6.8 Characteristics of Good Knowledge

- Accurate.
- Nonredundant.
- Consistent.
- As complete as possible (or at least reliable enough to draw conclusions).

6.9 Knowledge Acquisition Defined

Knowledge acquisition is the process by which knowledge available in the world is transformed into a representation that can be used by a system. It is multifaceted and encompasses many technical problems of knowledge engineering.

6.10 Knowledge Acquisition Steps

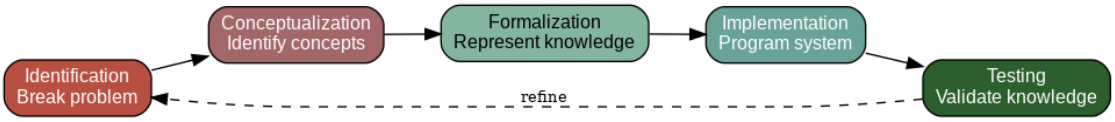


Diagram 13

1. **Identification** — break the problem into parts.
2. **Conceptualization** — identify concepts.
3. **Formalization** — represent knowledge.
4. **Implementation** — program the system.
5. **Testing** — validate the knowledge.

6.11 Knowledge Acquisition Methods

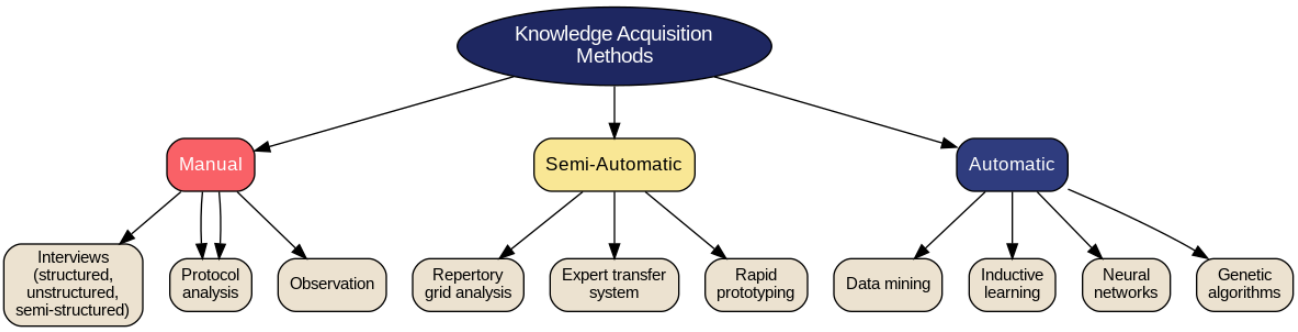


Diagram 14

Manual Methods

- **Interviews** — structured (goal-oriented, walk-through), unstructured (complex domains with unrelated data), or semi-structured.
- **Process tracking** — track reasoning processes.
- **Protocol analysis** — document expert decision-making; "think-around-process".
- **Observation** — motor and eye movements.
- **Other manual tools:** case analysis, critical incident, user discussions, expert commentary, graphs and conceptual models, brainstorming, prototyping, multidimensional scaling, clustering, iterative performance review.

Semi-Automatic Methods

- **Repertory grid analysis** — organized, perceptual model of expert knowledge; identifies domain objects and attributes; distinguishes objects via a grid.
- **Expert transfer system** — computer program that elicits information from experts.
- **Rapid prototyping** — determines sufficiency of available knowledge.

Computer-based tools typically feature: adding knowledge to a base, assessing/refining knowledge, visual modeling, decision-tree and rule creation, information-flow analysis, and integration tools.

Automatic Methods

- **Data mining** by computers.
- **Inductive learning** from recognized cases.
- **Neural computing** — mimicking the human brain.
- **Genetic algorithms** — using natural selection.

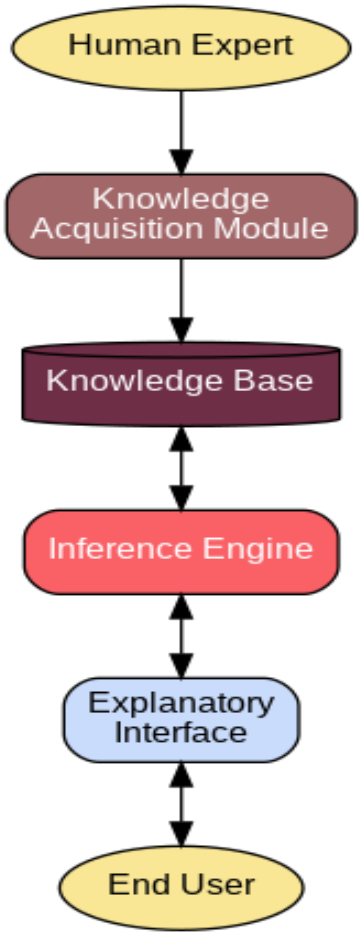
6.12 Knowledge Acquisition Difficulties

- Knowledge is not easy to acquire or maintain.
- Transferring knowledge between people is hard because of expression difficulties and the need to recast it into a form suitable for the knowledge base.
- Experts are busy or unwilling to share knowledge.
- Elicitation methods are not fully refined.
- Collection should involve several sources.
- Difficulty recognizing the relevant parts of the expert's knowledge.
- Experts change over time.

6.13 Types of Knowledge-Based Systems

- Expert Systems
- Neural Networks
- Case-Based Reasoning (CBR)
- Genetic Algorithms
- Intelligent Agents

Lecture 7 — Expert Systems and Case-Based Reasoning



7.1 What is an Expert?

An expert is an experienced practitioner — a highly effective problem-solver and decision-taker in a specific field.

Three qualities of experts:

1. They make good decisions.
2. They make those decisions quickly.
3. They cope with a wide range of problems.

7.2 Expert Systems

A knowledge-intensive program that captures the expertise of humans in a limited domain.

Reasons to build an expert system:

- Archive an expert's knowledge against their retirement or departure.
- Disseminate knowledge beyond the expert's physical location.
- Ensure uniformity of advice/decisions.
- Train other specialists.

7.3 Types of Expert Systems

Type	Description	Examples
Rule-based (Diagnostic) systems	IF–THEN statement-driven	Doctor, technician, car mechanic
Identification systems	Model of human knowledge	Materials spillage, bacterial agent identifier
Inference engines	Search through rule base — forward or backward chaining	Reasoning core of many systems
Decision Support Systems	Planning, scheduling, designing	Planning tools
Expert System Shell	Programming environment for an expert system	Development shells

7.4 Components of an Expert System

1. **Knowledge-based module** — stores knowledge in a particular representation.
2. **Inference engine** — uses the knowledge base to reach conclusions; must understand the KB format.
3. **Explanatory interface** — where the human interacts with the system.
4. **Knowledge acquisition module** — helps build new knowledge bases.

7.5 Advantages of Expert Systems

- Knowledge is **permanent** (doesn't leave with the expert).
- Knowledge is **easily replicated**.
- Knowledge is represented **explicitly** and can be evaluated.
- The system is **consistent** — unlike humans, computers don't have bad days.
- **Low running costs** once built.

7.6 Disadvantages of Expert Systems

- Developing an expert system is usually costly in time and money.
- Historically high failure rate, especially during the knowledge acquisition phase.
- Even successful development can fail at the organizational adoption stage.
- Human experts can update their knowledge via common sense, cross-domain knowledge, and contact with peers — expert systems cannot.

7.7 Building an Expert System

Two primary steps:

1. **Knowledge acquisition** — extracting knowledge and methods from the expert.
2. **Knowledge representation** — reforming the knowledge into an organized form usable by the system.

7.8 Case-Based Reasoning (CBR)

CBR uses a database of past cases and solutions.

Core workflow:

1. The user describes the problem.
2. The system searches the database for similar cases.
3. The system asks follow-up questions.
4. It finds the closest fit and retrieves the solution.
5. The solution is modified as required.
6. The new problem and solution are stored as a new case.

7.9 Steps in CBR

1. Collect the important features that define each new case.
2. Retrieve past cases matching those features most closely.
3. Select the cases most relevant to the current problem.
4. Adapt the stored cases to solve the current problem if necessary.
5. Validate the new solution and store it as a new case.
6. When no case match is found, locate an alternative solution and record both the problem and its solution.
7. When multiple matches exist:
   - Resolve ambiguities if multiple solutions appear.
   - Recognize that multiple solutions may sometimes be acceptable.

7.10 CBR Process (The 4 R's)

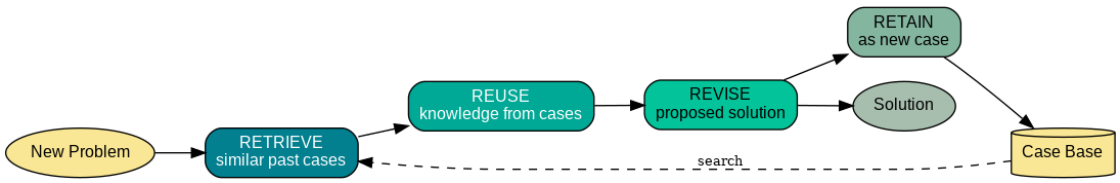


Diagram 16

A CBR system effectively follows processes or rules to:

- **Retrieve** the most relevant cases for comparison with the current problem.
- **Reuse** knowledge from those cases to help solve the current problem.
- **Revise** the proposed solution using case information.
- **Retain** details of the now-solved problem as a new case for future reference.

A key advantage: cases do not need to be understood by the knowledge engineer in order to be stored.

Reference

Turban, Aronson, and Liang. *Decision Support Systems and Intelligent Systems*, 7th Edition. Prentice Hall, 2005.

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